Smart Car Redesign & Autonomous Navigation

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AISE 4020
AI Systems Engineering Design I

Project Objectives

REDESIGN

Redesign the Freenove Smart Car for Improved Efficiency & Reliability

OPTIMIZE

Reduce Power Consumption & Simplify Wiring

VISION

Implement Vision-based Navigation & Ultrasonic Obstacle Detection

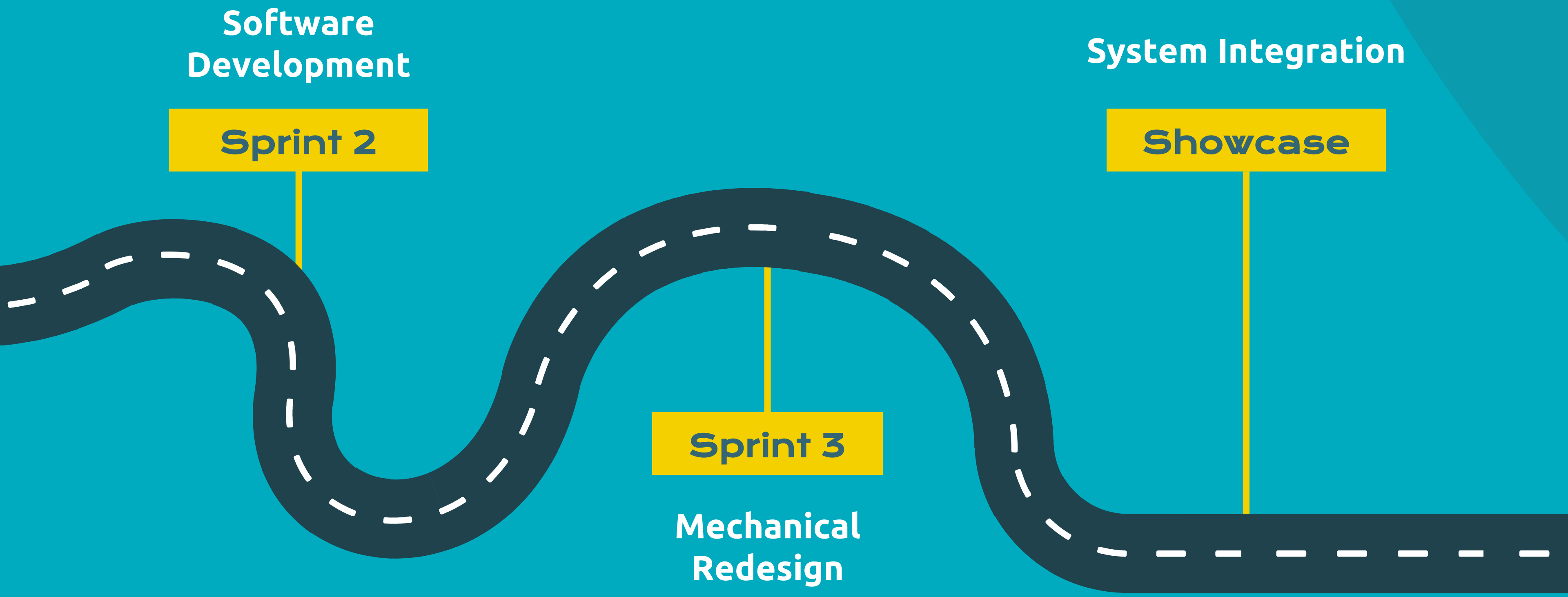
NAVIGATION

Integrate AprilTag Detection for Autonomous Navigation

MANUFACTURE

Develop a Stable & Manufacturable Mechanical Structure

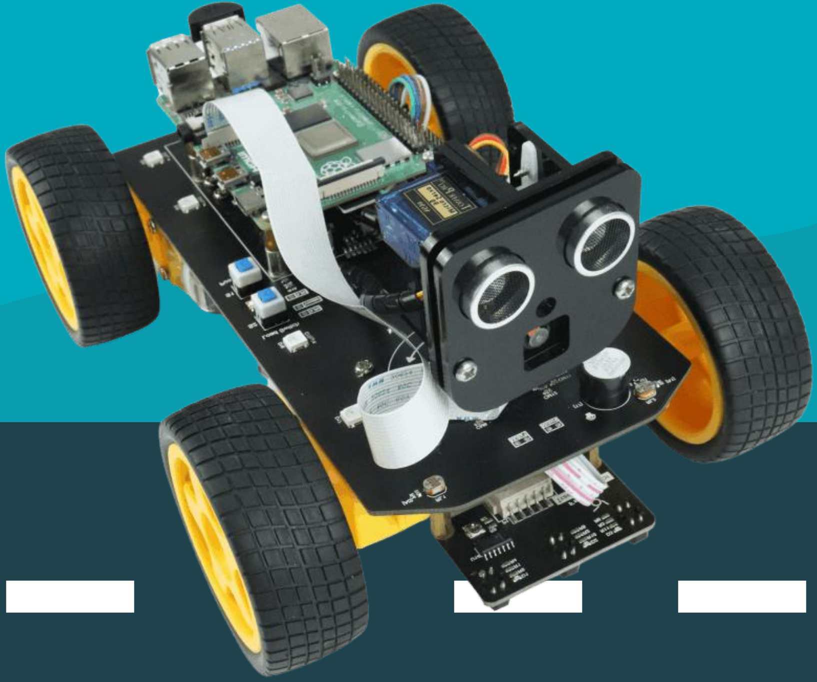
High-Level Overview



Redesign Comparision

4WD Smart Car

Our Solution



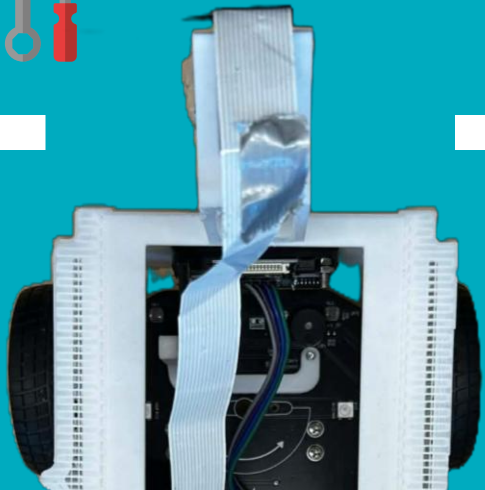
Comprehensive Breakdown

Mechanical Redesign

- Converted 4WD → Front-Wheel Tank Drive
- Rear Ball Mount for Stability
- Custom Durable & Lightweight Chassis
- Fixed Camera Mount
- Reduced Wiring & Component Count

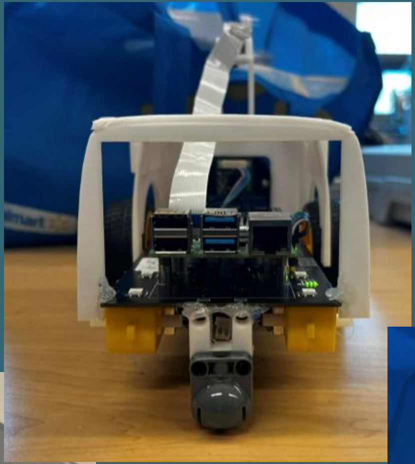
Software & Vision

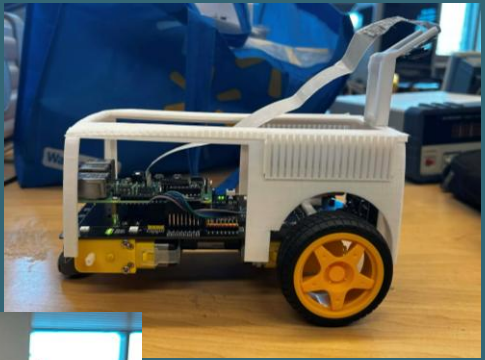
- Grayscale Image Processing
- Obstacle Detection with Ultrasonic
- AprilTag Detection with OpenCV
- State-Based Motion Control
- Tank Drive Motor Control Logic

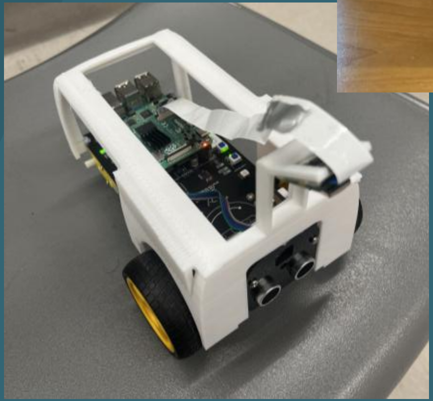


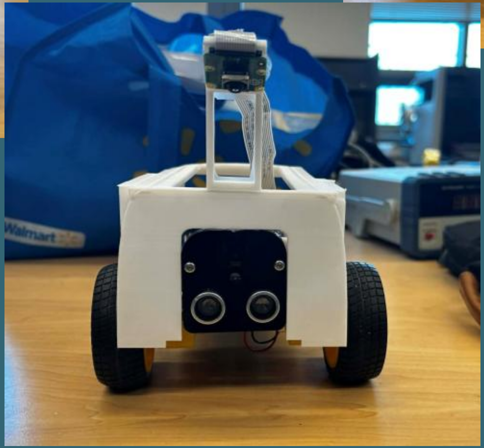

LIVE DEMO


KD
KD








Walmart

QUESTIONS & COMMENTS?

MSE-02